

System Architecture, Codebase Structure & Technical Manual

A Comprehensive Engineering Guide to the Full-Stack Agritech Platform for Crop Disease Diagnostics, Epidemiological Risk Forecasting, and Agricultural Ecosystem Integration.

Repository: Zinbas/Cropshield_9	Environment: Node 22+ / React 19 / tRPC 11
Target Domain: Agritech / Computer Vision / Microclimate	Database: Drizzle ORM (PostgreSQL / TiDB / MySQL)
AI Vision Gateway: Google Gemini Multimodal	Weather API: Open-Meteo REST Service
Production Deployment: Vercel Serverless Platform	Document Date: September 2026

1. Executive Summary & Core Capabilities

CropShield is an enterprise-ready, full-stack agritech platform specifically engineered to bridge the gap between smallholder farmers, agricultural extension workers, certified agronomists, and local input retailers. The application operates as a hybrid offline-first digital assistant that combines computer vision plant pathology with real-time microclimate epidemiological forecasting.

Key Core Capabilities:

- Multimodal Crop Health Scanning:** Farmers capture or upload field photographs of crops. The system integrates optional agronomic context (soil type, soil pH, moisture, acreage, and field observations) with local weather metrics and routes them to Google Gemini for structured, conservative diagnostic assessments.
- Microclimate Epidemiology Engine:** A rule-based and predictive risk engine calculates disease and pest vulnerabilities based on real-time temperature, humidity, precipitation, crop growth stage, and regional disease patterns over 7 to 15-day horizons.
- Interactive Regional Risk Heatmap:** Administrators and field officers monitor high-risk outbreaks through aggregated regional clusters without compromising individual farmer geolocation privacy.
- Follow-Up Cases & Actionable Treatment Tracking:** Scan findings automatically generate persistent case records with step-by-step treatment checklists that farmers can check off over time.
- Verified Agricultural Ecosystem Directory:** Direct geocoded access to certified extension experts and licensed agrochemical/seed supply stores with one-tap dialing and messaging.

Key Insight: CropShield strictly treats AI diagnostic outputs as clinical guidance rather than authoritative guarantees. The system enforces conservative confidence bounds and urges expert verification for high and critical threats.

2. System Architecture & Technology Stack

The application is structured as a unified monorepo with strict layer boundaries, end-to-end type safety, and isolated development and production execution pipelines.

Layer	Technologies Employed	Architectural Role & Scope
Frontend Client	React 19, TypeScript, Vite 5, Tailwind CSS v4, Wouter	Single-page app with olive-green design system, responsive mobile navigation bar, skeleton loaders, and multilingual i18n support (English, Hindi, Marathi, Assamese, Bengali).
UI & UX Forms	Radix UI, Framer Motion, React Hook Form, Zod	Accessible unstyled primitives (Radix), fluid animations (Framer), and strictly-typed robust form validation matching backend Zod schemas.
Data Visualization & Maps	Recharts, Google Maps API	Charts for analytics distribution and interactive risk heatmaps / GPS-assisted geolocation for precise user/profile mapping.
Backend API	Node.js, Express 4, tRPC 11, Zod v4, SuperJSON, Jose JWT	End-to-end type-safe RPC API, JWT cookie sessions, local & OAuth auth providers, role-based authorization guards.
Database / ORM	Drizzle ORM, PostgreSQL / MySQL	Type-safe database abstraction covering 11 tables, enum definitions, relational queries, and migration versioning.
AI Vision Layer	Google Gemini Multimodal (gemini-3.6-flash / configurable)	Structured JSON schema extraction for symptoms, crop classification, pathogen determination, and tailored treatment plans.
Meteorological API	Open-Meteo Weather REST API	Real-time temperature, humidity, rainfall, wind velocity, and 7-day daily forecasts with resilient fallback states.
Media Storage	@aws-sdk/client-s3, Local Disk Fallback	Decoupled binary storage keeps database records lean; images stored locally or via S3 presigned URLs (Forge credentials).
Testing Strategy	Vitest, Playwright E2E	Automated unit suites for auth/risk/persistence contracts and browser-driven end-to-end flows for admin workflows.

3. Complete Repository Directory & File Map

CropShield organizes all frontend, backend, database, shared modules, and operational guides into an intuitive, maintainable monorepo structure:

```
CropShield_9/
├── backend/
│   ├── _core/ # Framework infrastructure & runtime engine
│   ├── app.ts # Express base app, middlewares, trRPC mounting
│   ├── context.ts # Request context, user authentication extraction
│   ├── cookies.ts # Cross-environment secure cookie helpers
│   ├── env.ts # Environment variable parsing and validation
│   ├── index.ts # Local development server endpoint & port resolver
│   ├── llm.ts # Multi-provider LLM adapter (Gemini / Forge)
│   ├── notification.ts # Internal notification dispatchers
│   ├── oauth.ts # Third-party OAuth integration endpoints
│   ├── static.ts # Production static file serving
│   ├── storageProxy.ts # Media stream proxy for protected assets
│   ├── trpc.ts # trRPC procedure builders (public, protected, admin)
│   ├── vite.ts # Vite development server HMR middleware
│   ├── db.ts # Drizzle client, relational query methods, DB helpers
│   ├── localAuth.ts # Email/password authentication, JWT, session cookies
│   ├── riskEngine.ts # Epidemiological calculation rules & threat modeling
│   ├── routers.ts # Comprehensive trRPC procedure definitions
│   ├── seedTestData.ts # High-density realistic seed data generator
│   ├── storage.ts # S3 presigned URL client & local disk fallback
│   ├── *.test.ts # Vitest unit & contract suites (auth, risk, persistence)
├── frontend/
│   ├── public/ # Static assets, SVG icons, local upload targets
│   ├── src/
│   │   ├── _core/ # Core client hooks (auth, network states)
│   │   ├── components/ # Reusable feature & layout UI components
│   │   ├── ui/ # Radix UI primitives with Tailwind v4 styling
│   │   ├── AIChatBox.tsx # Interactive AI agronomy assistant dialog
│   │   ├── AdminOutbreakPanel.tsx # Outbreak management & officer escalation
│   │   ├── AnalyticsCharts.tsx # Risk distribution & trend visualizations
│   │   ├── DashboardLayout.tsx # Responsive shell with olive-theme nav
│   │   ├── InteractiveRiskMap.tsx # Geospatial outbreak & threat visualizer
│   │   ├── Map.tsx # Google Maps integration with tile fallback
│   │   ├── RiskPredictionPanel.tsx # 7-15 day disease/pest forecast cards
│   │   ├── SkeletonLoader.tsx # Tactile placeholder loaders
│   │   ├── contexts/ # ThemeContext (dark/light), AuthProvider
│   │   ├── hooks/ # Custom reactive hooks (useMobile, useDebounce)
│   │   ├── lib/ # Utility helpers (i18n, regionalRisk, trpc client)
│   │   ├── pages/ # Primary route views (Home.tsx, NotFound.tsx)
│   │   ├── App.tsx # Wouter router declaration & providers
│   │   ├── index.css # Global Tailwind tokens, olive-theme styles
│   │   ├── main.tsx # React 19 root bootstrap
│   ├── database/
│   │   ├── schema.ts # Drizzle table definitions, pgEnums, schema types
│   │   ├── *.sql # Incremental SQL migration scripts
│   ├── common/ # Shared TypeScript constants, error codes, and contracts
│   ├── docs/ # Operational runbooks, handover guides, test specs
│   ├── drizzle.config.ts # Drizzle Kit migration generator configuration
│   ├── package.json # Project dependencies, workspace scripts, engine locks
│   ├── tsconfig.json # Strict TypeScript compiler options & path aliases
│   ├── vite.config.ts # Vite bundler plugins, dev proxy, JSX transforms
```

4. Database Architecture & Schema Models

CropShield defines 11 distinct relational entities using Drizzle ORM. The schema supports strict relational integrity, role enforcement, and spatial coordinate tracking across districts and states.

Table Name	Primary Columns & Types	Business Purpose & Relationships
users	id (PK), openId (UK), name, email, loginMethod, passwordHash, role, accountStatus, timestamps	Core authentication record. Roles: 'user' (farmer) or 'admin'. Supports disabling accounts.
profiles	userId (FK), displayName, phone, state, district, pinCode, village, town, primaryCrop, experienceYears, lat/long, networkMode	Farmer agronomic identity, contact info, geolocation coordinates for microclimate weather retrieval. Supports distinct village & town persistence.
crops	id (PK), ownerId (FK), name, cropType, region, acreage, status ('healthy' 'monitoring' 'at_risk')	Farmer field plots. Connects scans and risk predictions to specific crops and acreages.
scans	id (PK), ownerId, cropId, imageKey, imageUrl, riskLevel, confidence, disease, symptoms, assessment, recommendations, recommendationProgress, soilType, soilPh, soilMoisture, cropCount, landArea, fieldNotes	Primary diagnostic record. Stores original photo reference, AI clinical output, optional field context (soil, area), and per-scan user progress checklist.
cases	id (PK), ownerId, scanId (UK FK), reference, status ('open' 'reviewing' 'resolved'), notes	Follow-up case tracking. Links a critical or high-risk scan to ongoing agronomic remediation.
experts	name, qualification, specialization, organization, experienceYears, state, district, lat/long, status ('pending' 'verified' 'rejected')	Verified extension officer directory. Geocoded to match nearby farmers for field visits or phone triage.
drugStores	name, ownerContact, phone, email, address, state, district, pinCode, lat/long, categories, licenseInfo, status	Licensed agricultural input retailers offering fertilizers, biological agents, and approved pesticides.
weatherCache	state, district, latitude, longitude, payload (JSON), fetchedAt, expiresAt	Caches Open-Meteo responses to minimize third-party API latency and handle poor network environments.
riskPredictions	ownerId, cropId, riskScore (0-100), riskLevel, threatType, threatDetails (JSON), weatherSnapshot, growthStage, validUntil, dismissed	Forecasted disease and pest vulnerabilities generated by the risk engine. Valid for 10-day outlooks.
riskAlertHistory	predictionId (FK), ownerId, alertType ('push' 'in_app' 'sms'), readAt, actionTaken, feedbackRating (1-5), feedbackNotes	Tracks farmer alert interaction, actions taken, and ground-truth user feedback on forecast accuracy.
regionalOutbreaks	state, district, threatType, reportCount, averageRiskScore, outbreakLevel ('watch' 'warning' 'outbreak'), officerNotified	District-level aggregate outbreak tracker. Triggers automated escalations to agricultural departments.

5. Deep Dive: Core Platform Engines & Workflows

5.1 Multimodal Crop Scan & Diagnostic Pipeline

The scanning pipeline in `backend/routers.ts` (`farmer.analyzeScan`) guarantees that every scan is transactionally recorded and resilient against external AI latency:

- Payload Validation & Sanitization:** Incoming images are checked for MIME type (JPEG/PNG/WebP), decoded from base64, and verified to be under the 12 MB limit.
- Pre-Analysis Persistence:** Before invoking the vision model, image bytes are written to storage (S3 or `frontend/public/uploads`), and an initial scan record is created with status 'analyzing'. This prevents data loss if the model connection drops.
- Agronomic Context Injection:** The system bundles farmer-provided field metrics (soil pH, moisture, crop count, land area) and live Open-Meteo weather into the prompt.
- Structured Schema Extraction:** The Gemini model is prompted with a strict JSON schema requiring: `cropType`, `riskLevel` (low, medium, high, critical), `confidence` (0-100), `symptoms[]`, `assessment`, `disease`, and `actionableRecommendations[]`.
- Progress Checklist Initialization:** Recommendations are transformed into an interactive checklist (`{[step, completed: false]}`) stored in `recommendationProgress`.
- High-Risk Event Dispatch:** If risk is high or critical, owner notifications are triggered, and the case becomes eligible for admin escalation.

5.2 Microclimate Epidemiological Risk Engine

Implemented in `backend/riskEngine.ts`, this engine combines weather physics, crop taxonomy, and phenology:

Vulnerability Factor	Meteorological / Agronomic Threshold	Pathological Impact
Extreme Humidity	Relative humidity >= 85% with temp 20°C - 30°C	Fungal spore germination (Powdery Mildew, Rust, Leaf Spot). Contributes +45 to fungal risk score.
Rainfall & Wet Canopy	Precipitation >= 5mm with high moisture	Bacterial blight, splash dispersal, damping-off. Contributes +35 to bacterial score.
Heat & Stagnant Air	Temp >= 32°C, humidity < 50%, wind < 10 km/h	Accelerated insect reproduction (Aphids, Thrips, Whiteflies). Contributes +40 to insect score.
Crop Growth Stage	Multipliers: Seedling (1.3x), Flowering (1.4x), Harvest (0.8x)	Adjusts baseline scores; flowering crops exhibit maximum sensitivity to insect and fungal invasion.

5.3 Regional Outbreak Intelligence & Administrator Heatmap

The administrative dashboard provides macro-level surveillance across districts. When multiple high-risk scans or threat predictions emerge in a district, `regionalOutbreaks` transitions from 'watch' to 'warning' or 'outbreak'. Administrators can trigger the `admin.escalateOutbreak` mutation, which dispatches emergency advisories to regional agricultural officers.

6. End-to-End Type-Safe API Specification (tRPC)

CropShield eliminates runtime API serialization mismatches by utilizing tRPC 11. Procedures are strictly partitioned across 6 distinct router namespaces:

Namespace	Procedure Name	Access	Description
auth	me	Public	Returns active session user from HTTP-only JWT cookie.
auth	signup / signin	Public	Registers/authenticates user, creates profile, sets session token.
auth	logout	Public	Clears local and OAuth authentication cookies.
farmer	snapshot	Protected	Returns aggregated farmer state: profile, crops, scans, active cases.
farmer	analyzeScan	Protected	Uploads crop photo, runs Gemini AI diagnosis, creates scan record.
farmer	updateRecommendationProgress	Protected	Updates completed checklist items for ongoing treatment.
farmer	reportThreat	Protected	Submits community threat report to regional outbreak tracker.
farmer	verifiedExperts / approvedDrugStores	Protected	Returns nearby verified experts and licensed farm supply stores.
weather	current	Public	Fetches live Open-Meteo metrics with 8-second circuit breaker.
risk	predict	Protected	Runs riskEngine algorithm and persists 10-day threat predictions.
risk	active / history	Protected	Queries non-dismissed alerts and historical prediction archive.
risk	feedback	Protected	Records farmer ground-truth rating (1-5) and actions taken.
admin	overview / directory	Admin Only	Returns platform stats, approved scans, and farmer directories.
admin	approveScan / setFarmerStatus	Admin Only	Moderates crop scans and manages farmer account active states.
admin	escalateOutbreak / resolveOutbreak	Admin Only	Notifies agriculture officers or marks regional outbreak resolved.
admin	seedTestData	Admin Only	Seeds realistic showcase farmers, scans, experts, and stores.

7. Local Development, Operations & Deployment

CropShield is designed for rapid onboarding and frictionless deployment across bare-metal, containers, or Vercel.

Step 1: Install Dependencies & Prepare Environment

Prerequisites: **Node.js 22+** and **pnpm 10+**.

```
# Install dependencies
pnpm install

# Configure Environment Variables (or export in shell)
# DATABASE_URL="mysql://user:pass@host:port/cropshield"
# GEMINI_API_KEY="your-gemini-api-key"
# JWT_SECRET="your-secure-jwt-secret"
# PORT=3000

# Push Database Migrations
pnpm db:push

# Launch Unified Development Server (Backend + Vite HMR)
pnpm dev
```

Step 2: Quality Assurance & Validation Commands

Before committing or promoting builds, execute the continuous integration pipeline:

```
# 1. Type-check TypeScript codebase without emit
pnpm check

# 2. Execute Vitest test suite (Auth, Risk Engine, Persistence Contracts)
pnpm test

# 3. Compile production bundles (Vite client + bundled Express server)
pnpm build

# 4. Run Playwright End-to-End test suite (Chromium headless)
pnpm test:e2e
```

Step 3: Production Deployment Workflow (Vercel)

The repository includes a dedicated `api/index.js` serverless adapter and a custom `vercel.json` build configuration. Pushes to the main branch automatically trigger Vercel preview builds, which can be promoted directly to the production alias (`cropshield-6.vercel.app`).